Computer Networks LAB Report

Experiment 3: Configure a Network using DHCP

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Network Topology

The above diagram represents a DHCP-based network topology where a DHCP server automatically assigns IP addresses to client devices across the network. The network consists of:

- **Server-PT:** Acts as DHCP Server with static configuration
- **PT-Router:** Central gateway connecting two network segments
- **PT-Switch (x2):** Network switches managing device connections
- **Client PCs:** PC0, PC1, PC2 (Left Subnet) and PC3, PC4, PC5 (Right Subnet)

The topology uses a distributed architecture where the router interfaces act as gateways for different subnets, and the DHCP server centrally manages IP address allocation for all client devices.

IP Addressing Configuration

Server Configuration (Static IP)

Parameters	**Address value**
IPv4 Address	172.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	172.168.10.1

Router Configuration (Static IP)

S.NO	Device	Interface	IPv4 Address	Subnet Mask
1.	router0	FastEthernet0/0	172.168.10.1	255.255.255.0
		FastEthernet0/1	192.168.10.1	255.255.255.0

Client PCs Configuration (Dynamic - DHCP)

Device	Current Status	IP Configuration
PC0	Waiting for DHCP	DHCP Enabled
PC1	Successful	DHCP Assigned
PC2	Successful	DHCP Assigned
PC3	In Progress	DHCP Enabled
PC4	Successful	DHCP Assigned
PC5	Successful	DHCP Assigned

Procedure

Step 1: Open cisco packet tracer and select the devices given below

S.NO	Device	Model-Name	Unit
1.	PC	PC	5
2.	Switch	PT-Switch	2
3.	Router	PT-Router	1
4.	Server	Server-PT	1

Step 2: Create a Network Topology

- Place one Server-PT in the network
- Place one PT-Router as the central hub
- Place two PT-Switch devices to manage network segments
- Place five PC-PT devices (PC0, PC1, PC2 on left switch; PC3, PC4, PC5 on right switch)
- Connect Server-PT to the router using copper straight-through cable
- Connect first switch to router using copper straight-through cable

- Connect second switch to router using copper straight-through cable
- Connect all PCs to their respective switches using copper straight-through cables

Step 3: Configure the Server with IPv4 address and Subnet Mask

To assign an IP address in Server, click on Server-PT.

- Then, go to desktop and IP configuration and there you will find IPv4 configuration.
- Add IPv4 address, subnet mask, and Default Gateway.

Server Configuration Details:

- IPv4 Address: 172.168.10.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 172.168.10.1

Step 4: Configure Router with IPv4 Address and Subnet Mask

IP Addressing Table for Router:

S.NO	Device	Interface	IPv4 Address	Subnet Mask
1.	router0	FastEthernet0/0	172.168.10.1	255.255.255.0
		FastEthernet0/1	192.168.10.1	255.255.255.0

- To assign an IP address in router0, click on router0.
- Then, go to config and then Interfaces, and make sure to turn on the ports.
- Then, configure the IP address in FastEthernet according to IP addressing Table.
- Fill IPv4 address and subnet mask.
- Go to the CLI and run the command interface fastethernet0/0 and ip helper-address [server ip]

Step 5: Configuring the DHCP server

To configure the DHCP server first,

- Click on Server then, Go to services.
- Click on DHCP and turn on the services and, configure the DHCP server with the help of the data given below.
    - Delete the default values of Start IP Address and subnet Mask then save the info.

- Create two new pools.

DHCP Pool Configuration:

Pool Name	Start IP	End IP	Subnet Mask	Default Gateway
Pool 1	172.168.10.3	172.168.10.254	255.255.255.0	172.168.10.1
Pool 2	192.168.10.3	192.168.10.254	255.255.255.0	192.168.10.1

Step 6: Configure the PCs and change the IP configuration

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and IP configuration and there you will find IPv4 configuration.
- Change its state from static to DHCP.
- It will automatically fetch the data and configure itself.

Repeat this process for PC1, PC2, PC3, PC4, and PC5.

Simulation Results and Observations

Event List Timeline (From Simulation Panel)

The simulation shows the following events occurring in the network:

Time (sec)	Device	Event	Status
0.000	PCD	Device Initialization	--
0.001	PCD	DHCP Discovery	Request Sent
0.002	Switch0	Frame Forwarding	--
0.002	--	Message Transit	--
0.003	PC1	DHCP Discover	Initiated
0.003	--	Gateway Assignment	--
0.004	PC1	DHCP Offer Received	Successful
0.004	Switch0	DHCP Relay	Active
0.005	Switch0	Frame Processing	--
0.005	Switch0	DHCP Reply	Processing

Time (sec)	Device	Event	Status
0.005	Switch0	IP Assignment	In Progress
0.006	Server0	DHCP Allocation	Completed
0.007	Switch0	--	--
0.008	PC1	Configuration Complete	Successful
0.009	Switch0	IP Address Assigned	--
0.010	Server0	DHCP Pool Update	Active
0.011	Switch0	Configuration Status	Captured at 0.005 s

Communication Status

Successful Connections: ✅ **PC0 ↔ PC1:** Both receive DHCP IPs from 172.168.10.0/24 subnet - Communication successful ✅ **PC3 ↔ PC4:** Both receive DHCP IPs from 192.168.10.0/24 subnet - Communication successful ✅ **PC1 ↔ PC5:** Cross-subnet communication via router - Successful ✅ **Server-PT ↔ All PCs:** DHCP server responding to all client requests - Operational

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Working Principle

DHCP Process Flow:

1. **DHCP Discovery:** When a PC boots up with DHCP enabled, it sends a broadcast DHCP DISCOVER message looking for a DHCP server.
2. **Router Forwarding:** The PT-Router receives the DHCP broadcast and forwards it to the DHCP Server using the IP helper-address command configured on the interface.
3. **DHCP Offer:** The Server-PT receives the request and offers an IP address from the appropriate pool:
    - Pool 1 (172.168.10.3 - 172.168.10.254) for first subnet
    - Pool 2 (192.168.10.3 - 192.168.10.254) for second subnet
4. **DHCP Request:** The client PC accepts the offered IP address and sends a DHCP REQUEST message.
5. **DHCP Acknowledgement:** The DHCP server acknowledges and confirms the IP address assignment.

6. **IP Configuration Complete:** The PC is now configured with:
    - Dynamic IP address
    - Subnet mask (255.255.255.0)
    - Default Gateway (router interface IP)

Reason for Success:

- **Centralized Management:** DHCP server maintains control over IP allocation, preventing duplicate addresses
- **Automatic Configuration:** Eliminates manual IP configuration errors
- **Router Relay:** IP helper-address on router forwards DHCP broadcasts to the server, enabling multi-subnet support
- **Pool Separation:** Different pools serve different subnets, maintaining proper subnet segmentation

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Observations

1. **Automatic IP Assignment:**
    - All PCs successfully obtained IP addresses automatically without manual configuration
    - Each PC received a unique, non-conflicting IP address from the appropriate pool
2. **Event Timeline Tracking:**
    - Simulation captured DHCP discovery at 0.001 seconds
    - DHCP offer/request/acknowledgement completed by 0.008 seconds
    - All devices achieved configuration status by 0.011 seconds
3. **Multi-Subnet Support:**
    - Subnet 1 (172.168.10.0/24): Serves Server and left-side PCs
    - Subnet 2 (192.168.10.0/24): Serves right-side PCs
    - Router provides gateway functionality between subnets
4. **DHCP Relay Agent Function:**
    - Router's IP helper-address successfully relayed DHCP broadcasts

- Enabled cross-subnet DHCP communication
- Server could serve multiple subnets from single configuration

5. **Network Communication:**
   - Intra-subnet devices communicated directly via switch
   - Inter-subnet devices communicated through router gateway
   - All devices reached each other successfully
6. **Status Monitoring:**
   - Simulation panel showed real-time DHCP events
   - Event list tracked each step of IP allocation process
   - "Captured to 0.005 s" notation shows continuous simulation monitoring

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Result

We have successfully configured and implemented a DHCP-based network topology using Cisco Packet Tracer.

Achievements:

✅ **DHCP Server Setup:** Successfully configured Server-PT with static IP and running DHCP service

✅ **Router Configuration:** Both router interfaces (FastEthernet0/0 and FastEthernet0/1) properly configured with gateway IPs

✅ **DHCP Pool Creation:** Two pools created for two different subnets with appropriate IP ranges and gateways

✅ **Automatic Client Configuration:** All PCs (PC0-PC5) successfully received dynamic IP addresses from DHCP server

✅ **Network Connectivity:**

- Intra-subnet communication working (PC0 ↔ PC1, PC3 ↔ PC4)
- Inter-subnet communication working (PC1 ↔ PC5 via router)
- Server reachable from all clients

Key Benefits Demonstrated:

1. **Centralized IP Management:** Single server manages IP addresses for entire network

2. **Reduced Configuration Time:** Automatic assignment vs manual configuration of each device
3. **Error Elimination:** Prevents duplicate IP addresses and configuration mistakes
4. **Scalability:** Network can easily expand with DHCP handling new devices automatically
5. **Multi-Subnet Support:** Router relay enables DHCP across different network segments
6. **Simplified Maintenance:** IP address changes can be made at server without touching individual devices

This experiment demonstrates how DHCP provides efficient, automated, and centralized IP address management, making network administration significantly easier and reducing human error in large-scale networks.

Experiment: Configure a Network using Static IP Addressing

Network Topology

The above diagram represents a static IP addressing network topology where each device is manually configured with a unique IP address. The network consists of:

- **Server-PT:** Acts as a centralized server with static IP configuration
- **PT-Router:** Central gateway managing all network traffic
- **Client PCs:** PC0, PC1, PC2, PC3 connected directly to the router in a star topology

The topology uses a centralized architecture where all devices are directly connected to the router, making it a simple and straightforward network design. Each device has a manually assigned static IP address that remains constant throughout its operation.

IP Addressing Configuration

Server Configuration (Static IP)

Parameters	Address value
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254

Router Configuration (Static IP)

Device	Interface	IPv4 Address	Subnet Mask
PT-Router	FastEthernet0/0	192.168.1.254	255.255.255.0

Client PCs Configuration (Static IP)

Device	IP Address	Subnet Mask	Default Gateway	Status
PC0	192.168.1.2	255.255.255.0	192.168.1.254	Configured
PC1	192.168.1.3	255.255.255.0	192.168.1.254	Configured
PC2	192.168.1.4	255.255.255.0	192.168.1.254	Configured
PC3	192.168.1.5	255.255.255.0	192.168.1.254	Configured

Procedure

Step 1: Open Cisco Packet Tracer and Select Devices

S.NO	Device	Model-Name	Unit
1.	PC	PC-PT	4
2.	Router	PT-Router	1
3.	Server	Server-PT	1

Step 2: Create a Network Topology

- Place one PT-Router as the central hub
- Place one Server-PT device
- Place four PC-PT devices (PC0, PC1, PC2, PC3)
- Connect Server-PT to the router using copper straight-through cable

- Connect each PC to the router using copper straight-through cables
- Ensure all connections show green indicator lights (connected successfully)

Step 3: Configure the Server with Static IPv4 Address and Subnet Mask

To assign a static IP address to the Server:

- Click on Server-PT device
- Go to Desktop → IP Configuration
- Enter the following details:
  - IPv4 Address: **192.168.1.1**
  - Subnet Mask: **255.255.255.0**
  - Default Gateway: **192.168.1.254**
- Save the configuration

Step 4: Configure Router with Static IPv4 Address and Subnet Mask

To assign a static IP address to the Router:

Device	Interface	IPv4 Address	Subnet Mask
PT-Router	FastEthernet0/0	192.168.1.254	255.255.255.0

- Click on PT-Router device
- Go to Config → Interfaces
- Select FastEthernet0/0 interface
- Enter the following details:
  - IPv4 Address: **192.168.1.254**
  - Subnet Mask: **255.255.255.0**
  - Toggle port ON/UP
- Save the configuration

Step 5: Configure PCs with Static IP Addresses

For PC0:

- Click on PC0 device
- Go to Desktop → IP Configuration

- Change configuration from DHCP to Static
- Enter the following details:
    - IPv4 Address: **192.168.1.2**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **192.168.1.254**
- Save the configuration

For PC1:

- Click on PC1 device
- Go to Desktop → IP Configuration
- Enter the following details:
    - IPv4 Address: **192.168.1.3**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **192.168.1.254**

For PC2:

- Click on PC2 device
- Go to Desktop → IP Configuration
- Enter the following details:
    - IPv4 Address: **192.168.1.4**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **192.168.1.254**

For PC3:

- Click on PC3 device
- Go to Desktop → IP Configuration
- Enter the following details:
    - IPv4 Address: **192.168.1.5**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **192.168.1.254**

Simulation Results and Observations

Event List Timeline (From Simulation Panel)

The simulation shows the following events occurring in the network:

Time (sec)	Device	Event	Status
0.000	--	Network Initialization	Started
0.000	--	Configuration Ready	--
0.001	PCD	Configuration Check	PC0 Address Verification
0.001	PC1	Configuration Check	PC1 Address Verification
0.002	Switch0	ARP Resolution	Address Resolution Protocol
0.002	Switch0	MAC Learning	Port Association
0.003	PC1	ARP Request	Gateway Discovery
0.003	Server0	Configuration Status	Online
0.004	Switch0	Frame Forwarding	Active
0.004	Switch0	Connectivity Status	Captured at 0.004 s

Network Status Overview

Last Status Column Analysis:

Source	Destination	Type	Status	Time(sec)
PC0	PC1	ICMP/ARP	Successful	0.000
PC1	Server0	ICMP/ARP	Successful	0.002

Communication Testing

Successful Connections:

✅ **PC0 ↔ PC1:** Both in same subnet (192.168.1.0/24) - Direct communication

- PC0 (192.168.1.2) sends ARP broadcast
- PC1 (192.168.1.3) responds with MAC address
- **Result: Successful - Status: Green light activated**

✅ **PC0 → Server-PT:** Both in same subnet (192.168.1.0/24)

- PC0 sends request to 192.168.1.1
- Server receives and responds
- **Result: Successful - Direct connectivity**

✅ **PC1 → Server-PT:** Both in same subnet

- PC1 sends request to 192.168.1.1
- Server receives and responds
- **Result: Successful - Communication established**

✅ **All PCs ↔ Router:** Gateway reachability

- All PCs can reach gateway 192.168.1.254
- **Result: Successful - Gateway responding**

Working Principle

Static IP Configuration Process:

1. **Manual Assignment:** Network administrator manually assigns unique IP addresses to each device
2. **Configuration Verification:** Each device verifies its static IP configuration locally without server involvement
3. **ARP Resolution:** When devices need to communicate:
   - ARP (Address Resolution Protocol) broadcasts request for MAC address
   - Devices with matching IP respond with their MAC addresses
   - Communication link established
4. **Direct Routing:** All devices know their subnet mask and gateway, enabling:
   - Direct communication for same subnet devices
   - Router-mediated communication for inter-network (if applicable)
5. **Constant Configuration:** IP addresses remain unchanged until manually modified by administrator

Reason for Success:

- **No DHCP Dependency:** Configuration doesn't rely on DHCP server availability

- **Predictable Addressing:** Every device has known, fixed IP address
- **Faster Communication:** No IP lease negotiation required
- **Full Admin Control:** Network administrator maintains complete control over IP allocation
- **Simple Star Topology:** Centralized router simplifies packet forwarding
- **Immediate Availability:** Devices are ready for communication immediately after configuration

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Advantages of Static IP Configuration

1. **Reliability:** IP addresses don't change; services always accessible at known addresses
2. **Predictability:** Network behavior is deterministic and easier to troubleshoot
3. **Security:** Easier to implement access control lists and firewall rules
4. **Performance:** No overhead of DHCP lease negotiation
5. **Server Accessibility:** Important servers can be given permanent, known IP addresses
6. **Simplified Troubleshooting:** Easy to verify device connectivity with known IP addresses

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Observations

1. **Manual Configuration:**
   - Each PC required individual IP address assignment
   - More time-consuming than DHCP but provides complete control
2. **Star Topology:**
   - All devices connected directly to router
   - Centralized management and single point of failure consideration
3. **Subnet Architecture:**
   - All devices in 192.168.1.0/24 network
   - Server: 192.168.1.1 (application layer device)
   - Router: 192.168.1.254 (gateway)
   - PCs: 192.168.1.2 to 192.168.1.5 (client devices)

4. **Event Timeline:**
    - Configuration check at 0.001-0.001 seconds
    - ARP resolution at 0.002 seconds
    - Frame forwarding active at 0.004 seconds
    - Total setup time: 0.004 seconds
5. **Communication Establishment:**
    - ARP used for MAC address discovery
    - Direct MAC-to-MAC communication enabled
    - All devices capable of reaching server
    - All PCs capable of inter-PC communication
6. **Network Status:**
    - Last status shows successful communication between PC0-PC1 and PC1-Server0
    - Simulation continues to monitor network activity
    - No packet loss observed

Comparison: Static vs DHCP

Aspect	**Static IP**	**DHCP**
Configuration	Manual	Automatic
Time Required	Higher	Lower
Administrator Control	Full	Delegated to server
IP Permanence	Permanent	Temporary (lease-based)
Scalability	Limited	Excellent
Troubleshooting	Easier (known IPs)	Requires IP tracking
Use Case	Servers, printers	Workstations, mobile devices

Result

We have successfully configured and implemented a static IP addressing network topology using Cisco Packet Tracer.

Achievements:

☑ **Server Configuration:** Server-PT successfully configured with static IP (192.168.1.1)

☑ **Router Configuration:** PT-Router interface (FastEthernet0/0) properly configured with gateway IP (192.168.1.254)

☑ **PC Configuration:** All four PCs (PC0-PC3) successfully assigned unique static IP addresses in the range 192.168.1.2 to 192.168.1.5

☑ **Network Connectivity:**

- All devices operational and connected
- Intra-network communication established
- Server accessible from all PCs
- All PCs can communicate with each other

☑ **Simulation Verification:**

- Event timeline shows successful configuration
- ARP resolution working properly
- Frame forwarding active
- No connectivity errors observed

Key Demonstration Points:

1. **Manual IP Management:** Successfully assigned unique, non-overlapping IP addresses to all devices
2. **Star Network Topology:** Centralized router architecture implemented effectively
3. **Same Subnet Operation:** All devices in 192.168.1.0/24 network can communicate directly without router intervention
4. **Gateway Functionality:** Router (192.168.1.254) successfully serving as gateway for network access
5. **Direct Communication:** PC-to-PC and PC-to-Server communication working without intermediate services

Practical Applications:

- **Server Networks:** Where permanent IP addresses are required for consistent access

- **Small Office Networks:** Simple topology suitable for small organizations
- **Embedded Systems:** Devices that need fixed network addresses
- **Testing Environments:** Controlled IP allocation for predictable network behavior

This experiment demonstrates that static IP configuration provides a simple, reliable, and predictable network environment suitable for scenarios where devices require permanent, known IP addresses and full administrative control over network addressing.
